

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

BE II (ME)

Roll No.

Kin. & Dyn. of Mac. (M)

SECOND B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(IV SEMESTER CBCS)

**4 ME - 43 A : KINEMATICS & DYNAMICS OF
MACHINES (M)**

Time - Three Hours

Maximum Marks - 80

- Note:-
- (i) Attempt any **FIVE** questions.
 - (ii) Answer should be to the point.
 - (iii) All questions carry equal marks.
 - (iv) Assume suitable data, if not given and mention it clearly.
 - (v) Draw neat sketches, wherever necessary.
 - (vi) Q.No. 1 complete solution (including calculations) is to be given in the drawing sheet.

1. A cam with 30 mm minimum radius is rotating clockwise at 1200 rpm to give the follower motion to a roller follower of 20 mm diameter. (i) Lift is 25 mm; (ii)

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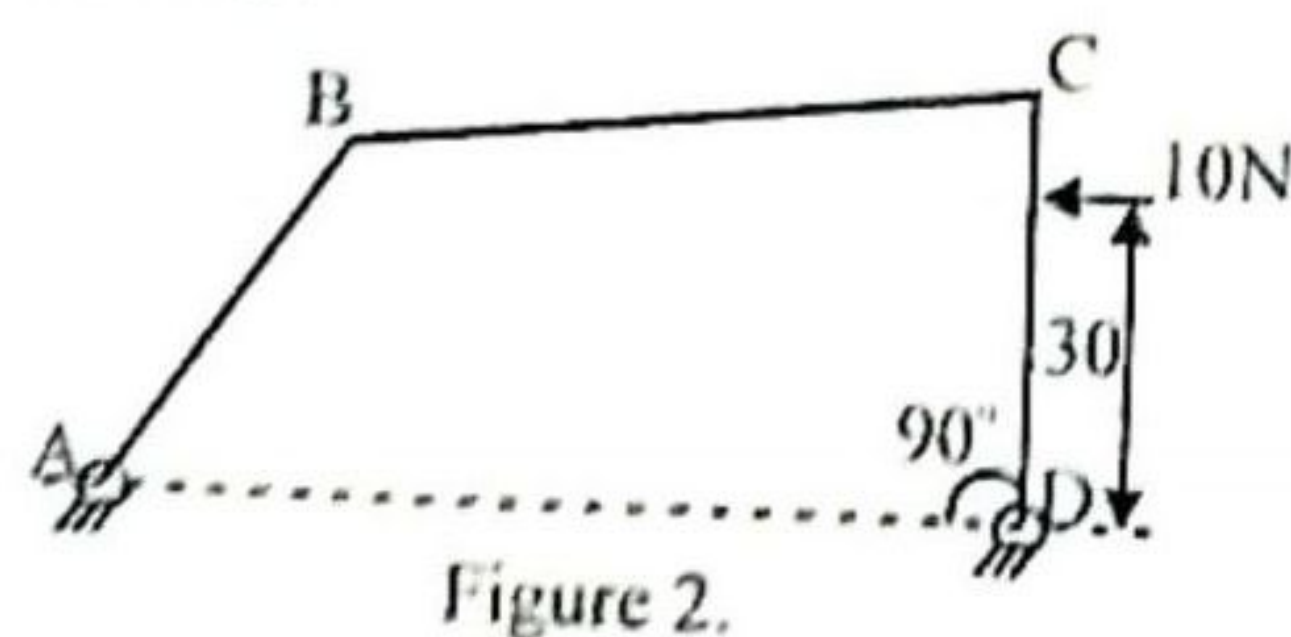
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5. (a) A motorcycle and its rider together weighs 2kN and their combined centre of gravity is 550 mm above the road when motorcycle is upright. Each wheel is of 580 mm diameter and has a moment of inertia of 1.0 kg-m^2 . The moment of inertia of rotating parts of engine is 0.15 kg-m^2 . The engine rotates at 5 times the speed of the vehicle and in the same sense. Determine the angle of heel necessary when motorcycle is taking a turn over a track of 35m radius at a speed of 60 mph. 8

- (b) Explain the effect of gyroscopic couple on a Naval Ship during (i) Steering (ii) Pitching and (iii) Rolling and draw neat diagrams for each case. 8

6. (a) An external force of 10 N is acting horizontally on the rocker link, 30 mm from the point D as shown in Figure 2. Find the amount of torque to be applied to the crank AB to keep the mechanism in static equilibrium and draw the free body diagram. Here, $AD = 80\text{mm}$, $AB = 30 \text{ mm}$, $BC = 70 \text{ mm}$, $CD=50 \text{ mm}$.



- (b) Explain the concept of free body diagrams using a suitable example. 6

7. Write short notes (any four). $4 \times 4 = 16$

- (a) Describe various types of cams and follower with neat sketches.
- (b) What is the function of a governor? How is it different from that of a flywheel?
- (c) Explain self - energising brakes.
- (d) What is the principle of virtual work? Explain.
- (e) Explain the gyroscopic effect on two-wheeled vehicles.